



Available online at www.sciencedirect.com

ScienceDirect

Journal of the Franklin Institute 359 (2022) 2788–2809

www.elsevier.com/locate/jfranklin



Autonomous dynamic docking of UAV based on UWB-vision in GPS-denied environment[☆]

Cheng Cheng^{a,b}, Xiuxian Li^{a,b,c,d}, Lihua Xie^e, Li Li^{a,b,c,*}

^a *Department of Control Science and Engineering, College of Electronics and Information Engineering, Tongji University, Shanghai 201804, China*

^b *Shanghai Institute of Intelligent Science and Technology, Tongji University, Shanghai 201804, China*

^c *Shanghai Research Institute for Intelligent Autonomous Systems, Shanghai, 201210, China*

^d *Institute for Advanced Study, Tongji University, Shanghai, 200092, China*

^e *School of Electrical and Electronic Engineering, Nanyang Technological University, 639798, Singapore*

Received 5 July 2021; received in revised form 11 January 2022; accepted 6 March 2022

Available online 17 March 2022

Abstract

This paper studies the autonomous docking between an Unmanned Aerial Vehicle (UAV) and a Mobile Platform (MP) based on UWB and vision sensors. To solve this problem, an integrated estimation and control scheme is proposed, which is divided into three phases: hovering, approaching and landing. In the hovering phase, the velocity of the MP and relative position between the MP and UAV are estimated by using geometric tools and Cayley-Menger determinant based on ultra-wideband distance measurements; in the approaching phase, a recursive least squares optimization algorithm with a forgetting factor is proposed, which uses distance, displacement and MP's velocity to estimate the relative position between the UAV and MP. With the estimated relative position, UAV can approach MP until reaching a distance such that MP is within the field of view of UAV; in the landing phase, the

[☆] This research was supported by the Shanghai Municipal Commission of Science and Technology No. 19511132100, 19511132101, the Shanghai Municipal Science and Technology Major Project, No. 2021SHZDZX0100, the National Natural Science Foundation of China under Grant 62003243, National Key R&D Program of China, No. 2018YFE0105000, 2018YFB1305304, and the Basic Science Centre Program by National Natural Science Foundation of China under grant 62088101.

* Corresponding author at: Department of Control Science and Engineering, College of Electronics and Information Engineering, Tongji University, Shanghai 201804, China.

E-mail addresses: fengyingcc@tongji.edu.cn (C. Cheng), xli@tongji.edu.cn (X. Li), elhxie@ntu.edu.sg (L. Xie), lili@tongji.edu.cn (L. Li).

UWB measurement value and visual perception attitude are integrated with the UAV on-board navigation sensor of the UAV to perform the precision landing. Simulation and experiment results verify the effectiveness and feasibility of the proposed integrated navigation scheme.

© 2022 The Franklin Institute. Published by Elsevier Ltd. All rights reserved.

1. Introduction

UAV autonomous docking has been a hot topic in recent years. Generally, it includes two stages: approaching phase and landing phase. First of all, UAV needs to locate the target relatively. The target can be a fixed landing point, manned or unmanned ground vehicles, or ships on the sea, etc. Secondly, the navigation is carried out according to the navigation rules when the relative position is determined. When the target enters the UAV's view of vision, the UAV will land on the set target area. Traditionally, because on-board vision system can provide rich and accurate position information, it is widely used in navigation and positioning [1–7]. Therefore, the landing problem is relatively mature using mainstream visualization methods, and most of the works are carried out in the approaching phase. In many cases, when GPS is available, the approaching capability can be achieved by simply transmitting GPS data between the UAV and target. For example, the obtained position and attitude in a multi-channel environment can be completed by RTK-GNSS technology [8]. Recent studies have shown that by integrating GPS and visual data, UAVs can land on vehicles with a speed of 50km/h [9], or fixed-wing UAVs can successfully dock with aerial targets [10]. However, this kind of method primarily depends on whether there is GPS coverage, and the accuracy also depends on the strength of the signal. In fact, GPS is not accessible in many places, such as indoor, canyon, forest, cave or deep well. As such, other methods have been proposed in the case of GPS-denied environments.

For example, [11] developed a model for autonomous navigation and collision avoidance of UAVs in a corridor environment with GPS constraints. Reference [12] proposed a localization method for a UAV, replacing the on-board GPS with noisy or unreliable GPS signals by aligning the images captured by a downward-facing monocular RGB camera with existing satellite images at flight positions. References [13,14] proposed an integrated relative position estimation and control scheme to achieve two objectives simultaneously: only using the distance and odometer measurements for relative positioning, and navigating the UAV to the required position by designing an appropriate bounded control law. Through a large number of outdoor flight tests, it is shown that UAV can approach a moving unmanned ground vehicle (UGV) from 50m away and then accurately land on a $1.5m \times 2m$ large platform.

Although the above works can complete the UAV autonomous landing task in many cases, it requires the vehicle to have GPS position information or vehicle odometer position information. Note that accurate odometry is generally difficult to obtain, especially for UAVs with low grade IMU. Therefore, this paper focuses on UAV docking on MP without the vehicle odometer in GPS-denied environments.

The main contents are as follows: an autonomous navigation framework is proposed, which is divided into three stages: hovering, approaching and landing. In the hovering stage, a speed and relative position estimation method is proposed based on distance measurement using geometric tools and extended Kalman filter (EKF). In the approaching stage, an estimation

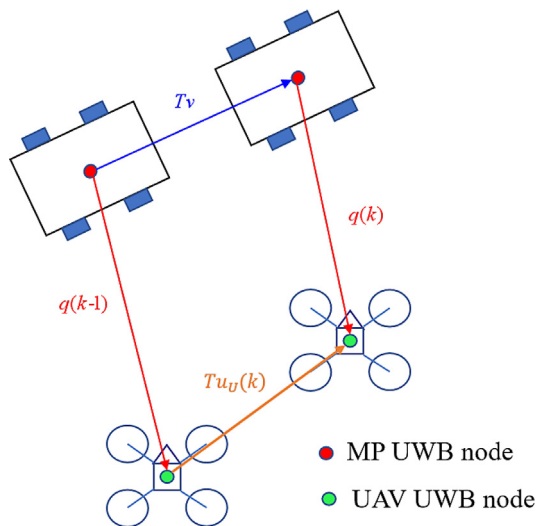


Fig. 1. Visualization of main concepts.

control law based on a forgetting least squares method is proposed, and its convergence is analyzed.

The main contributions of this paper are as follows. We design an autonomous dynamic navigation method based on UWB-vision sensors and propose a three stages approach to achieve UAV-MP docking in a GPS-denied environment. Relative localization is obtained only through UWB distance measurements, without the need of external infrastructure. Secondly, we design a method to estimate the relative position and MP velocity by measuring distance information, without MP carrying an on-board odometer. And after that, we develop an inverse proportional saturation navigation algorithm based on the estimated relative position. Finally, the proposed navigation algorithm is verified by simulation and experiment.

The rest of this paper is arranged as follows. In [Section 2](#), we introduce the basic problem definition and navigation framework. In [Section 3](#), the proposed integrated relative positioning and docking control scheme is detailed. In [Section 4](#), we report the simulation results. Finally, [Section 5](#) summarizes this paper.

2. System overview and problem formulation

In this section, we will refer directly to [Fig. 1](#) to define some basic concepts required in the following sections.

As there is a low-level controller that takes into consideration the UAV dynamics in tracking control, at the mission planning level, we assume a single integrator model:

$$p_U(k+1) = p_U(k) + Tu_U(k), \quad (1)$$

where $p_U(k) \in \mathbb{R}^3$ is the UAV position at time instant $k \geq 0$, T represents the sampling period, and $u_U(k)$ is the UAV control input. Similarly, we consider the MP dynamics as

$$p_M(k+1) = p_M(k) + Tv, \quad (2)$$

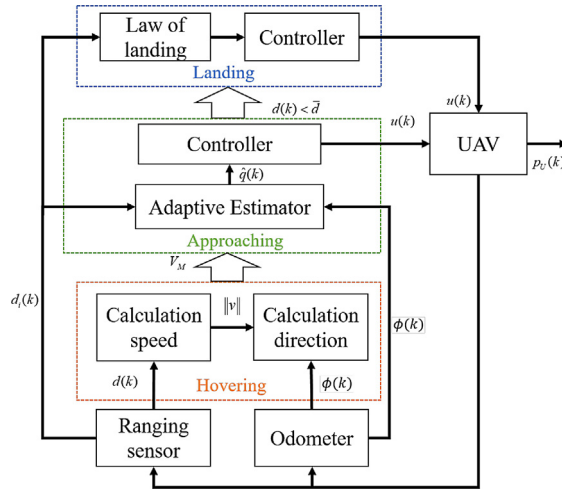


Fig. 2. Navigation and positioning control schemes.

where $p_M(k) \in \mathbb{R}^3$ is the MP position at time instant $k \geq 0$, and $v \in \mathbb{R}^3$ is a constant, that is, the MP is considered to move at a time-invariant velocity. In general, we assume that the MP is moving on the ground, so the height of the 3D parameter of MP is 0 and does not change.

Remark 1. It is worth noting that most of experiments in the literature are performed for the MP with a uniform linear motion [9,15]. Nevertheless, MP with non-zero acceleration is an interesting topic for future research.

In order for the MP to come within the field of camera view of the UAV, the position of the two needs to satisfy the following:

$$\|p_U - p_M\| \leq \bar{d}, \quad (3)$$

where $\bar{d} > 0$ is a constant, which depends on the altitude of the UAV and the maximal visual distance.

As shown in Fig. 1, we define the relative position of the UAV to the MP as follows,

$$q(k) \triangleq p_U(k) - p_M(k). \quad (4)$$

In this setup, a UWB node is installed on the MP, and another UWB node is installed on the UAV. We call the UWB node on the MP the anchor. It should be noted that the UAV node can only send information to locate one anchor point at the time.

The problem's objective is to estimate relative position of the MP based on distance measurement and displacement, to design navigation control algorithms, and to complete the autonomous docking.

As shown in Fig. 2, we propose an integrated navigation control scheme based on ranging sensor and odometer, which is divided into three phases: hovering, approaching and landing. $d(k)$ is the distance between the UAV and MP at time instant k by the UWB ranging sensor. $\phi(k)$ is the displacement of the UAV and measured by the-camera and IMU of the UAV, satisfying $\phi(k) = T u_U(k)$. In the hovering phase, the velocity of the MP is estimated by $d(k)$ and $\phi(k)$. In the approaching phase, navigation control is performed by estimating the

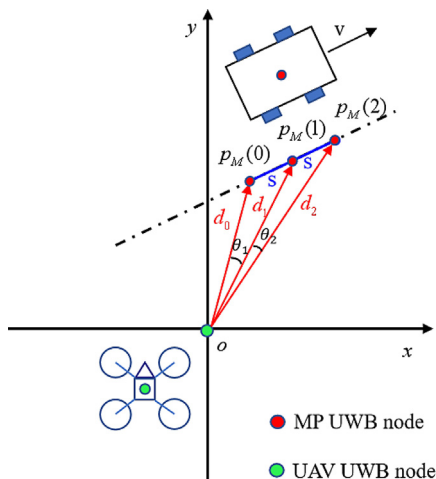


Fig. 3. Measurement of speed in hovering phase.

relative position based on the speed and distance information of the MP obtained in the previous phase. In the landing phase, the UAV makes a precise landing by means of the on-board vision sensors and UWB.

3. Integrated relative positioning and navigation algorithm

As mentioned above, the proposed autonomous landing process consists of three phases: hovering phase, approaching phase and landing control phase. In the hovering phase, assuming that the UAV hovers at altitude h at a point far away from the MP, the velocity and direction of MP and the relative position of the UAV to MP are calculated by the distance measurement based on UWB and the displacement of the UAV. In the approaching phase, a recursive least squares optimization algorithm based on adaptive genetic factors is proposed. The distance and relative displacement measurements are used to estimate the position of the UAV to the target. With this estimation, the UAV can approach MP. Once the UAV is located near MP and within the view of the UAV camera (i.e., landing phase), the system will switch to the camera-UWB-vision combination mode to ensure a high level of pose accuracy. In the final phase, vision-based landing is performed to complete advanced guided maneuvers to ensure a safe vertical landing on the mobile platform.

3.1. Estimation of the velocity of MP based on UWB

In the hovering phase, we measure the motion of MP by hovering the UAV, and then propose a velocity estimator based on Cayley-Menger determinant [16]. A schematic illustration is given in Fig. 3, where O is the hovering position of UAV, $p_M(k)$ is the position of MP at time k , d_i is the distance between UAV and MP measured by UWB ranging sensors, θ_k is the included angle of the lines between MP and UAV at time k and time $k-1$, and s is the moving distance for MP during each sampling time interval. Obviously, the speed $\|v\|$ of MP cannot be determined by only one distance measurement. In fact, it is known that it can be obtained by three distance measurements.

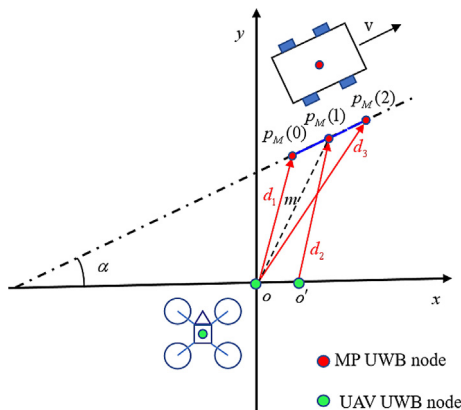


Fig. 4. Measurement of velocity direction in hovering phase.

It is easy to see that $s = T\|v\|$. For brevity, we define $x \triangleq s^2$, $l_0 \triangleq d_0^2$, $l_1 \triangleq d_1^2$, $l_2 \triangleq d_2^2$. According to the CayleyMenger determinant, we can obtain the algebraic equations describing the geometric relationship between x , d_0 , d_1 and d_2 , as follows.

$$\det \begin{vmatrix} 0 & x & 4x & l_0 & 1 \\ x & 0 & x & l_1 & 1 \\ 4x & x & 0 & l_2 & 1 \\ l_0 & l_1 & l_2 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 \end{vmatrix} = 0. \quad (5)$$

This equation has three solutions, $x_1 = 0$ and $x_2 = x_3 = \frac{l_0 - 2l_1 + l_2}{2}$. Since x has to be positive, then $s = \sqrt{\frac{d_0^2 - 2d_1^2 + d_2^2}{2}}$. Then one can obtain a unique solution for the speed of MP:

$$\|v\| = \frac{\sqrt{\frac{d_0^2 - 2d_1^2 + d_2^2}{2}}}{T}. \quad (6)$$

Remark 2. Note that the distance measurements from UWB sensors always contain noise. Therefore, we use EKF to estimate the measurement speed, and the feasibility will be verified in the following simulation section. Because this filter is widely used in various control fields [17–20], it will not be introduced here.

3.2. Calculation of MP velocity direction based on UWB measurement data

In order to determine the movement direction of the MP, the UAV cannot remain stationary and must perform some local movement. Here, we propose a possible maneuver strategy and continue with the discussion in the previous section, where distance measurements d_0 , d_1 and d_2 are performed at time 0, T and $2T$.

As shown in Fig. 4, we require the UAV to start linear motion at a given constant speed $\|v\|$ of MP at time 0, move from O point to O' , measure d_2 at time T , and then return to its initial position to make the planned measurement d_3 at time $2T$.

For simplicity, the UAV keeps the height constant from point O to point O' , while the height of the MP is known. In turn, the MP's velocity can be converted to a 2D plane calculation.

It should be noted that $\|v\|$ has been calculated by Eq. (6). Then we can calculate the distance $\|p_M(0) - p_M(1)\| = \|p_M(1) - p_M(2)\| = \|v\|T$. Now we check the distance between O , $p_M(0)$, $p_M(1)$ and $p_M(2)$. One can find that only $m \triangleq \|O - p_M(1)\|$ is not available. By solving the equations of CayleyMenger determinant of the corresponding four points,

$$D(O, p_M(0), p_M(1), p_M(2)) = 0, \quad (7)$$

where $D(O, p_M(0), p_M(1), p_M(2))$ is Cayley-Menger determinant of O , $p_M(0)$, $p_M(1)$ and $p_M(2)$, one can then get $m = \sqrt{\frac{d_1^2 + d_2^2}{2} - |v|^2 T^2}$. In order to find the direction of velocity v , we need to obtain the value of angle α as shown in Fig. 4. First of all, we notice that in a triangle, it can be concluded from geometric relations that

$$\alpha = \pi - \angle OO'p_M(1) - \angle p_M(0)p_M(1)O', \quad (8)$$

where $\angle OO'p_M(1)$ and $\angle p_M(0)p_M(1)O'$ can be calculated in $\triangle OO'p_M(1)$ and $\triangle p_M(0)p_M(1)O'$ by the law of Cosines.

Although obtaining the value of α , there is still a flip ambiguity for the direction of v with respect to the linear trajectory of the MPs movement. One simple way is to flight a triangle to remove this ambiguity. Therefore, the magnitude and direction of the MP velocity can be obtained by Eqs. (6) and (8).

Remark 3. Through the calculations in the first two sections, we get the velocity magnitude and direction of the MP, which, together with the distance measurements by the UWB sensors, enables us to determine the relative position $q(k)$. When measurement noises exist, we will use the EKF method to estimate the MP's velocity. Because the considered MP is assumed to perform a uniform linear motion, the velocity estimate can be rather accurate. It should be noted that most existing works focus on moving platforms with uniform speed and linear motion [9].

Remark 4. The direction angle of MP can be calculated by reducing the problem to a 2D plane as shown in Fig. 4. The height difference between UAV and MP anchor nodes is kept as h . $D(k)$ is expressed as the distance directly measured by the UWB sensor, and the distance between the projected point of the UAV on the 2D plane, and the MP is given $d(k) = \sqrt{D(k)^2 - h^2}$.

3.3. Adaptive estimator update law for relative localization

According to Fig. 2, we design a saturated bounded inverse proportional velocity controller $u_U(k)$ as

$$u_U(k) = -\frac{U_{\max}}{\max\{U_{\max}, G\|\hat{q}(k-1)\|\}} G\hat{q}(k-1), \quad (9)$$

where $U_{\max}$ is the maximum speed allowed for the UAV, $G > 0$ is the proportional gain, and $\hat{q}(k)$ denotes the estimate of the relative position $q(k)$. From Eq. (9), it can be observed that the controller can be calculated by the proportional gain and the estimate of relative positions.

By reviewing the definitions in the previous section and Fig. 1, we can verify the following relationships between $q(k)$ and the measured values $d(k)$, T , $u_U(k)$, v :

$$d^2(k) = [q(k-1) + Tu(k)]^\top [q(k-1) + Tu(k)], \quad (10)$$

$$d^2(k-1) = q^\top(k-1)q(k-1), \quad (11)$$

where $u(k) \triangleq u_U(k) - v$.

The following equation can be obtained by Eqs. (10) and (11):

$$d^2(k) - d^2(k-1) = T^2 \|u(k)\|^2 + 2Tu^\top(k)q(k-1). \quad (12)$$

Define

$$\zeta(k) \triangleq \frac{1}{2}(d(k)^2 - d^2(k-1) - T^2 \|u(k)\|^2), \quad (13)$$

$$\varphi(k) \triangleq Tu(k), \quad (14)$$

and by Eq. (12), one then has that

$$\zeta(k) = \varphi^\top(k)q(k-1). \quad (15)$$

For Eq. (15), in order to enhance the amount of information provided by new data and gradually weaken the old data to prevent data saturation, the cost function of $\hat{q}(k)$ is designed as:

$$J(k) = \sum_{i=1}^k \lambda^{k-i} [\zeta(i) - \varphi^\top(i)\hat{q}(i-1)]^2, \quad (16)$$

where $\lambda \in (0, 1)$ is the forgetting factor. Using the constructed estimation system Eq. (15), a forgetting factor least squares (FFLS) [21] based $\hat{q}(k)$ can be expressed as:

$$K(k) = P(k)\varphi(k) = \frac{P(k-1)\varphi(k)}{\lambda + \varphi^\top(k)P(k-1)\varphi(k)}, \quad (17a)$$

$$P(k) = \frac{1}{\lambda} [I - K(k)\varphi^\top(k)]P(k-1), \quad (17b)$$

$$\hat{q}(k) = \hat{q}(k-1) + K(k)[\zeta(k) - \varphi^\top(k)\hat{q}(k-1)], \quad (17c)$$

where $P(-1) = P_{-1}$, and $P(k)$, $\forall k \geq 0$ are $n \times n$ positive definite matrices [22,23]. From Ioannou and Fidan [22], Ioannou and Sun [23], it is known that

$$P^{-1}(k) = \lambda P^{-1}(k-1) + \varphi(k)\varphi^\top(k). \quad (18)$$

3.4. Convergence analysis

Assumption 1. The relative position changing rates $\omega(k) \triangleq q(k) - q(k-1)$ of UAV and MP is uncorrelated with $\omega(j)$ when $k \neq j$. Meanwhile, the changing rate should be square bounded, i.e., $\|\omega(k)\| < \sigma < \infty$, for some $\sigma > 0$.

Remark 5. For the changing rate of the relative position difference in Assumption 1, ω is uncorrelated and bounded because in practice there is always some error in the UAV control displacement.

Theorem 1. For systems Eqs. (1) and (2) with controller Eq. (9), under Assumption 1, if there exist $\alpha > 0$ and $\beta > 0$ such that $\varphi(k)$ satisfies the persistent excitation (PE) condition as follows:

$$\alpha I \leq \sum_{s=1}^N \varphi(k+s)\varphi^\top(k+s) \leq \beta I, \quad (19)$$

for any $k > 0$, where $N \geq 3$, $0 < \alpha \leq \beta < \infty$, then the $\hat{q}(k) - q(k)$ is bounded, i.e.,

$$\lim_{k \rightarrow \infty} \|\hat{q}(k) - q(k)\| \leq \frac{\beta\sigma}{\alpha} \left(\frac{N}{N-1} \right)^{N-1}. \quad (20)$$

Proof. Let

$$\tilde{q}(k) \triangleq \hat{q}(k) - q(k). \quad (21)$$

According to formulas Eqs. (16)–(20), it yields

$$\begin{aligned} \tilde{q}(k) &= \hat{q}(k) - q(k) \\ &= \hat{q}(k) - [q(k-1) + w(k)] \\ &= [I - P(k)\varphi(k)\varphi^\top(k)]\tilde{q}(k-1) - \omega(k) \\ &= \lambda P(k)P^{-1}(k)\tilde{q}(k-1) - \omega(k) \\ &= \lambda^k P(k)P^{-1}(0)\tilde{q}(0) - P(k) \sum_{i=1}^k \lambda^{k-i} P^{-1}(i)\omega(i). \end{aligned} \quad (22)$$

We define $\gamma_1(k) \triangleq \lambda^k P(k)P^{-1}(0)\tilde{q}(0)$ and $\gamma_2(k) \triangleq P(k) \sum_{i=1}^k \lambda^{k-i} P^{-1}(i)\omega(i)$.

From the definition of $P^{-1}(k)$, one can conclude that

$$\begin{aligned} P^{-1}(k) &= \lambda P^{-1}(k-1) + \varphi(k)\varphi^\top(k) \\ &= \lambda^k P^{-1}(0) + \sum_{i=1}^k \lambda^i \varphi(i)\varphi^\top(i) \\ &\leq \lambda^k P^{-1}(0) + \sum_{i=1}^k \lambda^i \left[\sum_{s=0}^{N-1} \varphi(i+s)\varphi^\top(i+s) \right]. \end{aligned} \quad (23)$$

According to the above equation, using the PE condition, it can be derived that

$$\begin{aligned} P^{-1}(k) &\leq \sum_{i=1}^k \lambda^i [\beta I] + \lambda^k P^{-1}(0) \\ &= \frac{1 - \lambda^k}{1 - \lambda} \beta I + \lambda^k P^{-1}(0) \\ &= \frac{\beta}{1 - \lambda} I + \lambda^k \left[P^{-1}(0) - \frac{\beta}{1 - \lambda} I \right]. \end{aligned} \quad (24)$$

Therefore, it follows that

$$\lim_{k \rightarrow \infty} P^{-1}(k) \leq \frac{\beta}{1 - \lambda} I. \quad (25)$$

Similarly, one has that

$$\frac{\lambda^{N-1} \alpha}{N(1 - \lambda)} I \leq \lim_{k \rightarrow \infty} P^{-1}(k). \quad (26)$$

It is not difficult to obtain that

$$\begin{aligned} \lim_{k \rightarrow \infty} \|\gamma_1(k)\| &= \lim_{k \rightarrow \infty} \lambda^k \|P(k)P^{-1}(0)\tilde{q}(0)\| \\ &\leq \lim_{k \rightarrow \infty} \frac{\lambda^k(1 - \lambda)}{\alpha N \lambda^{(N-1)}} \|P^{-1}(0)\|\|\tilde{q}(0)\| \\ &= 0, \end{aligned} \quad (27)$$

and

$$\begin{aligned} \|\gamma_2(k)\|^2 &= \left\| P(k) \sum_{i=1}^k \lambda^{k-i} P^{-1}(i) w(i) \right\|^2 \\ &\leq \frac{(1 - \lambda)^2}{\alpha^2 N^2 \lambda^{2(N-1)}} \left\| \sum_{i=1}^k \lambda^{k-i} P^{-1}(i) w(i) \right\|^2 \\ &= \frac{(1 - \lambda)^2}{\alpha^2 N^2 \lambda^{2(N-1)}} \cdot \left[\sum_{i=1}^k \sum_{j=1}^k \lambda^{2k-i-j} w^T(i) \cdot P^{-1}(i) P^{-1}(j) w(j) \right]. \end{aligned} \quad (28)$$

By using the formula $2x^T y \leq x^T x + y^T y$, one can get that

$$\begin{aligned} \lim_{k \rightarrow \infty} \|\gamma_2(k)\|^2 &\leq \frac{(1 - \lambda)^2}{\alpha^2 N^2 \lambda^{2(N-1)}} \cdot \lim_{k \rightarrow \infty} \sum_{i=1}^k \sum_{j=1}^k \lambda^{2k-i-j} \frac{1}{2} \cdot [w^T(i) P^{-2}(i) w(i) + w^T(j) P^{-2}(j) w(j)] \\ &\leq \lim_{k \rightarrow \infty} \frac{(1 - \lambda)^2}{\alpha^2 N^2 \lambda^{2(N-1)}} \sum_{i=1}^k \sum_{j=1}^k \lambda^{2k-i-j} \frac{\beta^2 \sigma^2}{(1 - \lambda)^2} \\ &= \lim_{k \rightarrow \infty} \frac{\beta^2 \sigma^2}{\alpha^2 N^2 \lambda^{2(N-1)}} \frac{(1 - \lambda^k)^2}{(1 - \lambda)^2} \\ &= \frac{\beta^2 \sigma^2}{\alpha^2 N^2 \lambda^{2(N-1)} (1 - \lambda)^2} \\ &\triangleq f(\lambda). \end{aligned} \quad (29)$$

By $f'(\lambda) = 0$, it yields $\lambda = \frac{N-1}{N}$, which, together with the above equation, results in

$$\lim_{k \rightarrow \infty} \|\gamma_2(k)\| \leq \frac{\beta \sigma}{\alpha} \left(\frac{N}{N-1} \right)^{N-1}. \quad (30)$$

Therefore, the conclusion of [Theorem 1](#) can be obtained by using [Eqs. \(27\) and \(30\)](#). $\square$

Remark 6. From [Eq. \(14\)](#), $\varphi(k)$ is determined by the displacement of the UAV and MP. The physical meaning of Assumption [Eq. \(19\)](#) is that the components of $\varphi(k)$, denoted by $\varphi_x(k)$,

$\varphi_y(k)$ and $\varphi_z(k)$, are linearly independent. Wherein, $\varphi_z(k)$ is always zero, so $\varphi(k)$ can be projected onto the 2D plane. Expanding the middle side of Eq. (19), one can obtain that

$$\sum_{s=1}^N \varphi(k+s) \varphi^\top(k+s) = \left[\sum_{s=1}^N \varphi_x(k)^2 \sum_{s=1}^N \varphi_x(k) \varphi_y(k) \sum_{s=1}^N \varphi_x(k) \varphi_y(k) \sum_{s=1}^N \varphi_y(k)^2 \right]. \quad (31)$$

Abiding by the CauchySchwarz inequality, one has that

$$\left(\sum_{s=1}^N \varphi_x(k) \varphi_y(k) \right)^2 \leq \sum_{s=1}^N (\varphi_x(k))^2 \sum_{s=1}^N (\varphi_y(k))^2, \quad (32)$$

for any $k > 0$. And the two sides of Eq. (32) are equal if and only if $\varphi_x(k)$ and $\varphi_y(k)$ are linearly dependent. As a result, Assumption Eq. (19) is equivalent to the linear independence of $\varphi_x(k)$ and $\varphi_y(k)$. Hence, it is reasonable to assume the PE condition, which is also widely employed in the literature [13–15,24].

Remark 7. In the GPS-denied environment, reference [12] proposed a map localization method based on a monocular camera for UAV by comparing captured images with pre-existing satellite images at the flight position. On the contrary, the method proposed in this paper does not need map information and training. References [13,14] proposed a scheme for docking by relative distance, displacement, and on-board displacement, in which the vehicle must carry an on-board odometer. In contrast, for the method proposed in this paper, the vehicle does not need to carry an odometer, which greatly saves docking costs and communication resources.

4. Simulation and experiment

4.1. Simulation

Hovering phase: we set the speed of the MP to be 5m/s for motion at (25,5), UAV is hovering at (0,0) for ranging, sampling period is $T = 0.1s$. We collect 50 range sample points and use the UWB sensor to measure the distance $d(k)$ between UAV and MP, its measurement error noise is random in the range of $(-0.1, 0.1)$. The estimates of the speed and error comparative analysis are shown in Figs. 5 and 6, where the observed value is calculated by (6) from a series of distance measurements, and the EKF- v value is calculated by EKF.

According to the data in Fig. 5, the speed $\|v\|$ estimated by EKF updating iteration is close to the real velocity. At the same time, Fig. 6 shows that the method proposed in this paper has a small error and good accuracy and can complete the calculation of MP velocity of UAV in hovering phase.

And then, we consider the speed of MP in maintaining uniform linear motion adding $\pm 0.1m/s$ of random noise error. The experimental results are shown in Figs. 7 and 8.

The simulation of the hovering phase shows that there are both sensor noise and moving platform speed noise, and the speed value estimated by EKF is within 0.5m from the real value of the MP.

Approaching phase: we design MP initial coordinates as (25, 5, 0.2), where 0.2 is the height between MP deck and ground. It is known that the initial hovering position of the UAV in

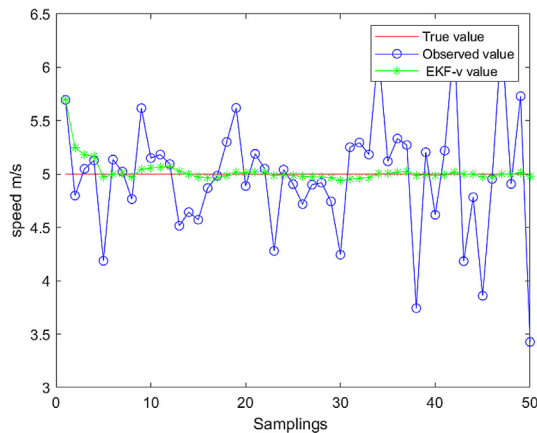


Fig. 5. Velocity $\|v\|$ results estimated by EKF. (Sensor measurement noise).

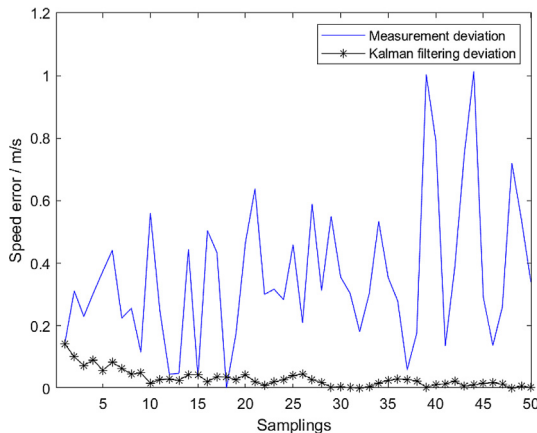


Fig. 6. Error comparison. (Sensor measurement noise).

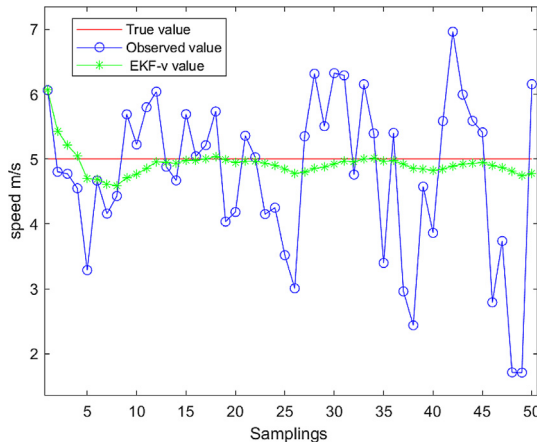


Fig. 7. Velocity $\|v\|$ results estimated by EKF. (Sensor measurement noise and MP's speed noise).

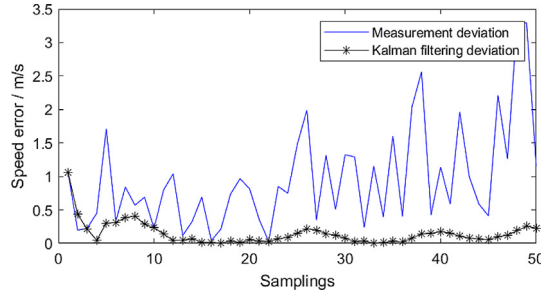


Fig. 8. Error comparison. (Sensor measurement noise and MP's speed noise).

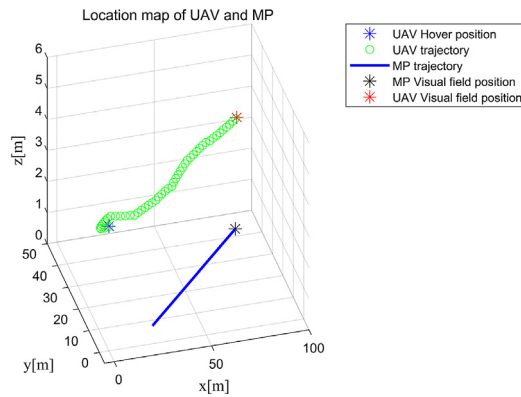


Fig. 9. Position of space motion trajectory. (Sensor measurement noise).

the previous stage is set at $(0, 0, 4)$, and 4 is the height of the UAV from the ground. From the hovering stage, we get the estimated speed and direction of MP. We set the real speed of MP as 5 m/s and the maximum speed of UAV as 8 m/s for simulation. The sampling period is $T = 0.1\text{ s}$, forgetting factor is $\lambda = 0.8$, and UAV control coefficient is $G = 0.85$. Assume that the UAV is at an altitude of 4 m . The wide-angle of the on-board camera is projected down to the ground as the center of a circle with a radius of 5 m (when the measured distance $d(k) \leq 5\text{ m}$, the MP is considered to be within the field of view of the UAV). A random noise of $(-0.1, 0.1)$ is added to the range sensor measurement (where the accuracy of the UWB anchor we selected is within $\pm 10\text{ cm}$). The results of the simulation are shown in Figs. 9–12.

As can be seen from Fig. 9, the UAV gradually approaches the MP using the control law proposed in this paper until the MP appears within the range of the UAV vision sensor and the range between the UAV and MP is within a preset value (where the height difference set in the simulation is 3.8 m). In addition, we can use visual landing. From the data in Fig. 10, we can see that our relative position, relative position estimation and relative position error will converge. The spatial coordinates of the vehicle and UAV are shown in Fig. 11. We can see that the UAV always approaches the MP in 2D coordinates, and finally reaches the top of the MP, so that UAV can make the landing. Fig. 12 shows the distance $d(k)$ which approaches to zero.

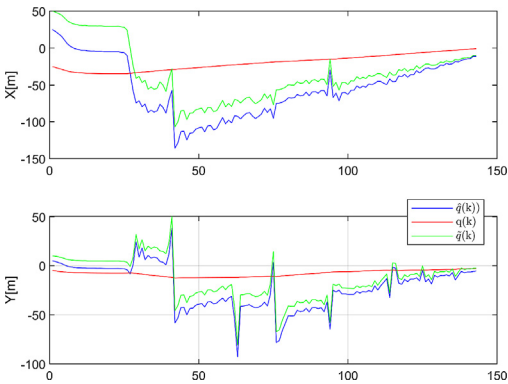


Fig. 10. Relative position estimation analysis chart. (Sensor measurement noise).

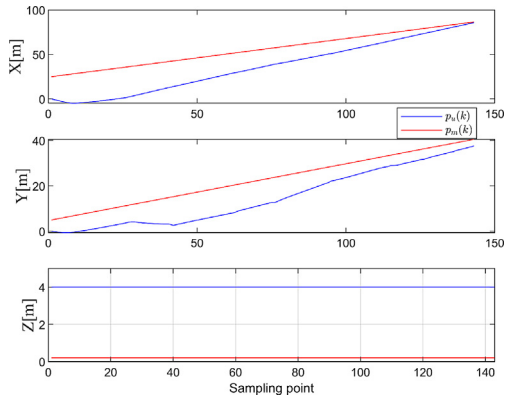


Fig. 11. UAV and MP space position. (Sensor measurement noise).

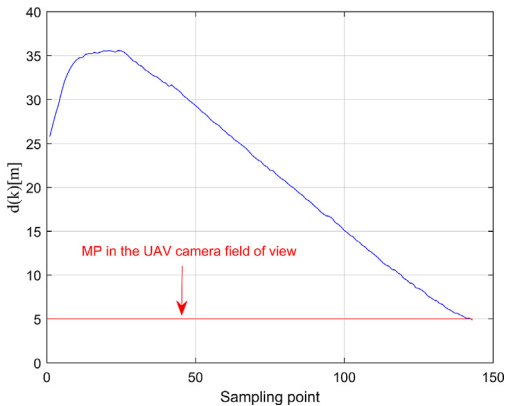


Fig. 12. UWB anchor measurement data. (Sensor measurement noise).

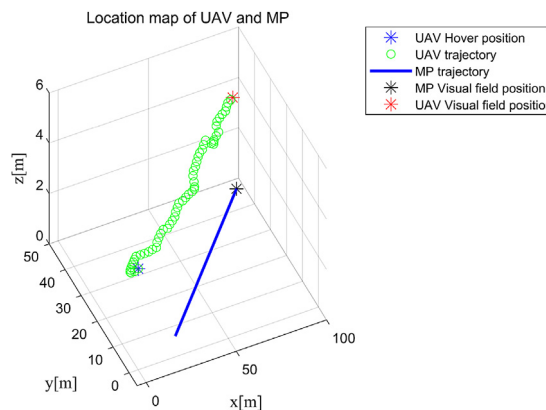


Fig. 13. Position of space motion trajectory. (Sensor measurement noise and MP's speed noise).

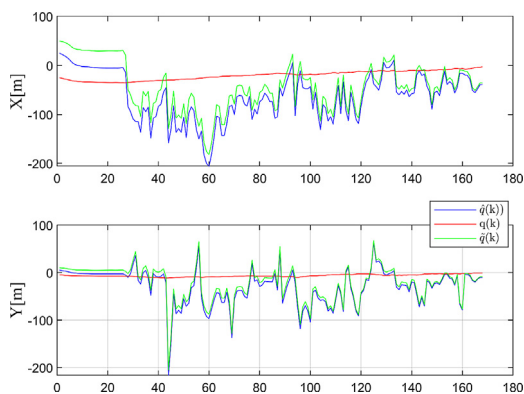


Fig. 14. Relative position estimation analysis chart. (Sensor measurement noise and MP's speed noise).

In order to further explore the anti-interference performance of the algorithm, a random noise of $\pm 0.1\text{m/s}$ is added to the speed of MP. The simulation results are shown in Figs. 13–16.

In conclusion, the simulation results show that the proposed algorithm can still complete the navigation task when there exist noises in both UWB measurements and MP velocity.

4.2. Experiment

Our system is implemented on a hexacopter equipped with a Pixhawk 4¹ flight controller, which also provides IMU measurements. Intel Realsense T265² camera is installed under the UAV to provide image information, which is processed by Intel NUC Core i7 Mini

¹ <http://px4.io>

² <https://www.intelrealsense.com/>

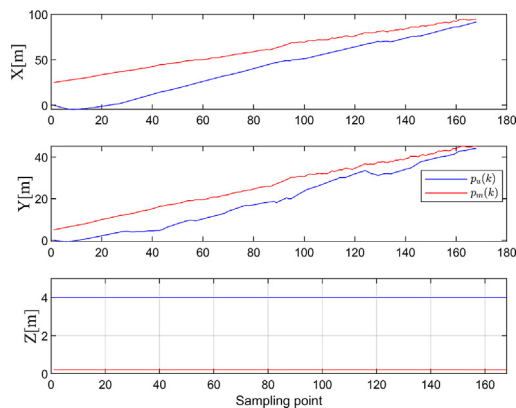


Fig. 15. UAV and MP space position. (Sensor measurement noise and MP's speed noise).

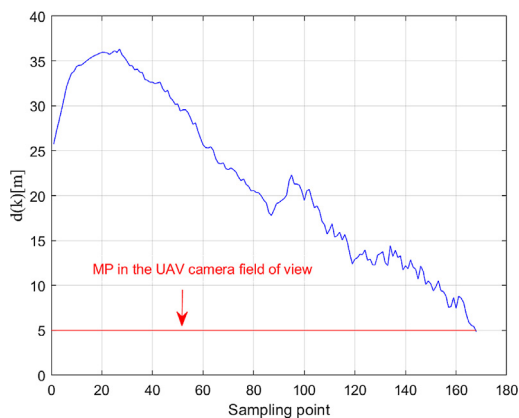


Fig. 16. UWB anchor measurement data. (Sensor measurement noise and MP's speed noise).

Computer. Distance measurements $d(k)$ are obtained by using two UWB nodes, which can provide accurate values with less than 10cm error within 100m range (DWM1000 module³). The MP is $1m \times 0.8m \times 0.3m$ consisting of 4 McNamee wheels, without any other sensing or positioning devices installed but with only one UWB-node on the deck. For the landing phase, we used AprilTag 2 technology [25] for the guided landing of the UAV. The specific working principle and installation are shown in Figs. 17–19.

Our experiment in some underground car park is artificially set up with a few feature points due to the lack of feature points on the smooth ground. The hovering height of the UAV is set to be 1.2m, the linear motion speed of the MP is 0.5m/s and the sampling period for data acquisition is 200ms. In the hovering phase, the UAV hovered to 1.2m, stayed for 5s to collect distance information, then moved 1m to the right and 1m forward to determine the MP velocity direction as well as to determine the initial relative position (flew a triangle to remove the ambiguity). The data obtained from the UWB sensor measurements, the MP

³ <https://www.decawave.com/product/dwm1000-module/>

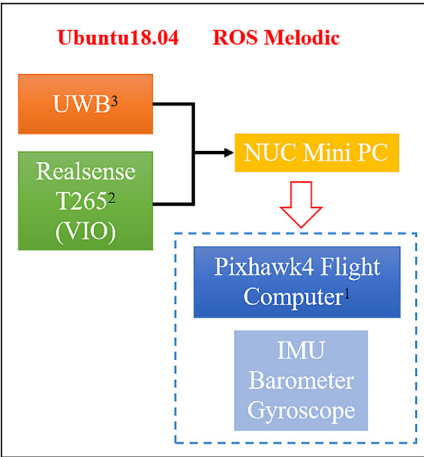


Fig. 17. Working principle.

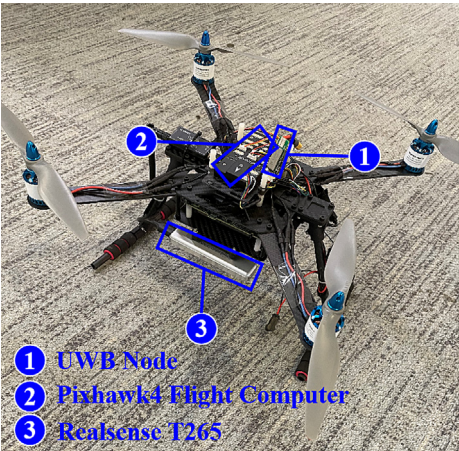


Fig. 18. Experiment setup. (UAV).

velocity calculated by (6) and the EKF estimate are shown in Fig. 20. During the approaching and landing phases, the UAV approached the MP through the proposed saturation control law, and after the Tag was detected, a visually guided landing is activated and the experiment is shown in Fig. 21. The trajectory of the UAV in the overall experiment obtained by ROS Rviz⁴ and QGC flight review⁵ is shown in Figs. 22 and 23. The video recording of the experiment can be found at <https://youtu.be/y3LC7biqEYA>.

The real-world experiment described above demonstrates the feasibility of the proposed scheme.

⁴ <http://wiki.ros.org/rviz/>
⁵ <https://logs.px4.io/>

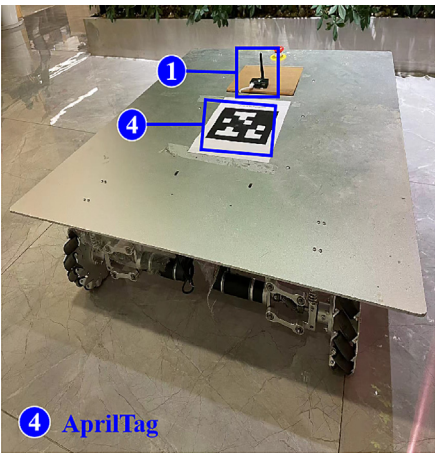
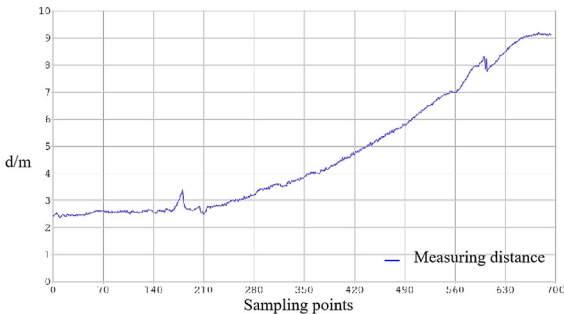
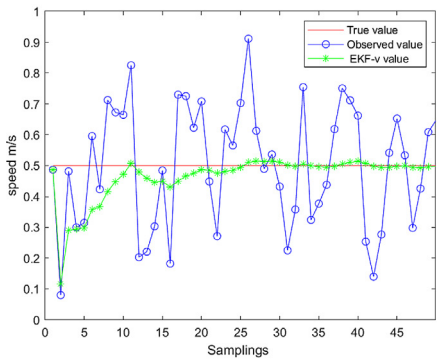


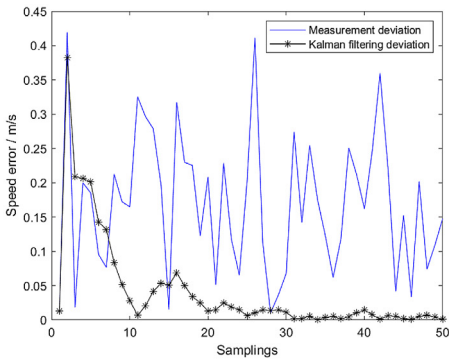
Fig. 19. Experiment setup. (MP).



(a) UWB measurement data

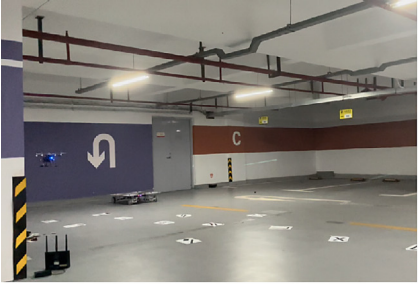


(b) MP's speed estimation

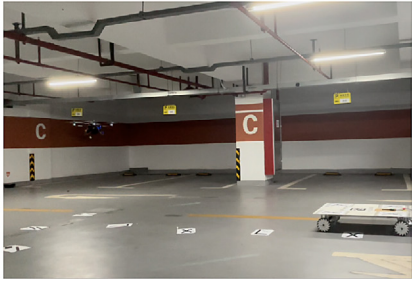


(c) Error analysis

Fig. 20. Velocity estimation in hovering phase.



(a) The hovering phase



(b) The approaching phase



(c) The landing phase

Fig. 21. Overall experimental process.

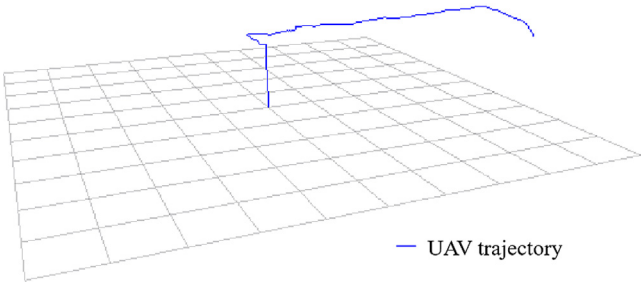


Fig. 22. The UAV trajectory visualization by Rviz.

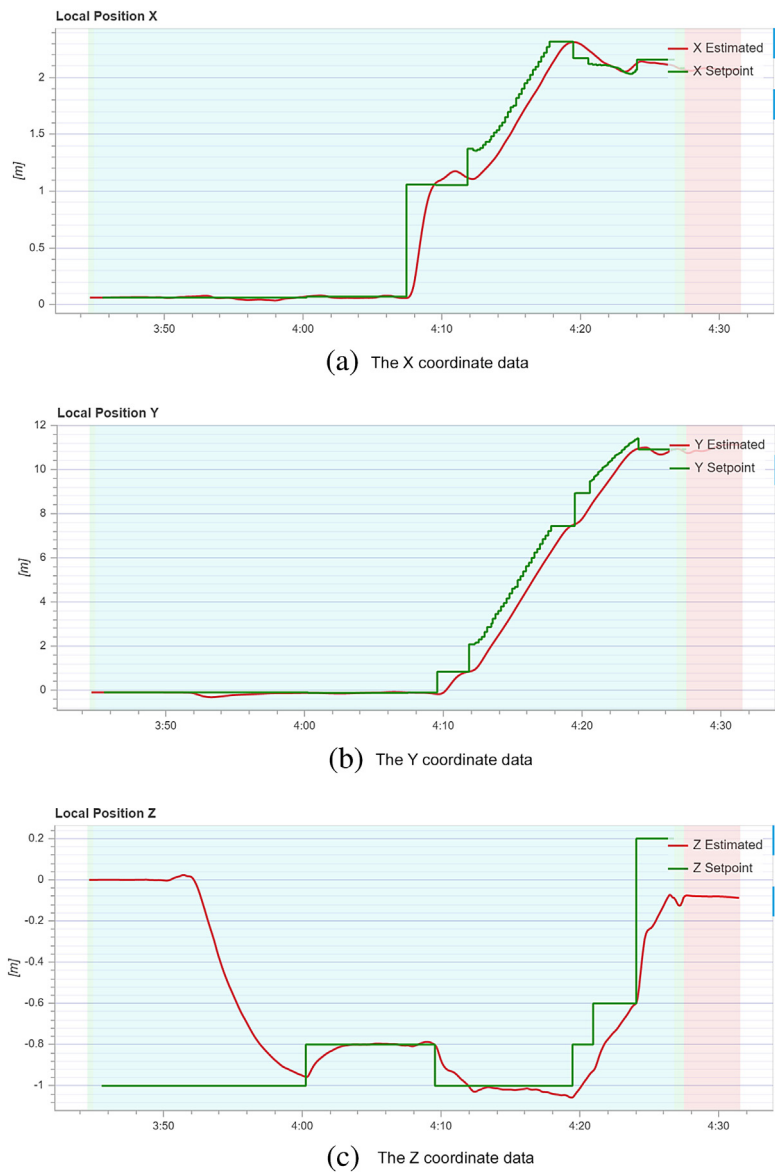


Fig. 23. The UAV flight path data analysis.

5. Conclusion

An autonomous docking scheme based on distance and displacement measurements was proposed for GPS-denied environments and MP without on-board odometers. We proposed an integrated relative positioning and navigation scheme to achieve autonomous docking. Simulation and experiment verified the effectiveness of the proposed docking scheme. The feasibility of the proposed scheme was verified in the GPS-denied environment.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- [1] T.B. Karamat, R.G. Lins, S.N. Givigi, A. Noureldin, Novel EKF-based vision/inertial system integration for improved navigation, *IEEE Trans. Instrum. Meas.* 67 (1) (2017) 116–125.
- [2] H. Cheng, T. Chen, C. Tien, Motion estimation by hybrid optical flow technology for UAV landing in an unvisited area (2019).
- [3] T. Hinzmann, T. Stastny, C. Cadena, R. Siegwart, I. Gilitschenski, Free LSD: prior-free visual landing site detection for autonomous planes, *IEEE Rob. Autom. Lett.* 3 (3) (2018) 2545–2552.
- [4] R. Polvara, S. Sharma, J. Wan, A. Manning, R. Sutton, Vision-based autonomous landing of a quadrotor on the perturbed deck of an unmanned surface vehicle, *Drones* 2 (2) (2018) 15.
- [5] L. Zhang, F. Deng, J. Chen, Y. Bi, S.K. Phang, X. Chen, B.M. Chen, Vision-based target three-dimensional geolocation using unmanned aerial vehicles, *IEEE Trans. Ind. Electron.* 65 (10) (2018) 8052–8061.
- [6] S. Zhong, P. Chirarattananon, Direct visual-inertial ego-motion estimation via iterated extended Kalman filter, *IEEE Rob. Autom. Lett.* 5 (2) (2020) 1476–1483.
- [7] W. Zhao, H. Liu, X. Wang, Robust visual servoing control for quadrotors landing on a moving target, *Journal of the Franklin Institute*, *J. Franklin Inst.* 4 (2) (2021) 2301–2319.
- [8] J.M. Hansen, T.A. Johansen, N. Sokolova, T.I. Fossen, Nonlinear observer for tightly coupled integrated inertial navigation aided by RTK-GNSS measurements, *IEEE Trans. Control Syst. Technol.* 27 (3) (2018) 1084–1099.
- [9] D.B. Wilson, A. Göktogan, S. Sukkarieh, Guidance and navigation for UAV airborne docking, in: *Robotics: Science and Systems*, vol. 3, 2015.
- [10] T. Muskardian, A. Coelho, E.R. Della Noce, A. Ollero, K. Kondak, Energy-based cooperative control for landing fixed-wing UAVs on mobile platforms under communication delays, *IEEE Rob. Autom. Lett.* 5 (4) (2020) 5081–5088.
- [11] R.P. Padhy, F. Xia, S.K. Choudhury, P.K. Sa, S. Bakshi, Monocular vision aided autonomous UAV navigation in indoor corridor environments, *IEEE Trans. Sustain. Comput.* 4 (1) (2018) 96–108.
- [12] H. Goforth, S. Lucey, GPS-denied UAV localization using pre-existing satellite imagery, in: *2019 International Conference on Robotics and Automation (ICRA)*, IEEE, 2019, pp. 2974–2980.
- [13] T.-M. Nguyen, T.H. Nguyen, M. Cao, Z. Qiu, L. Xie, Integrated UWB-vision approach for autonomous docking of UAVs in GPS-denied environments, in: *2019 International Conference on Robotics and Automation (ICRA)*, IEEE, 2019a, pp. 9603–9609.
- [14] T.-M. Nguyen, Z. Qiu, M. Cao, T.H. Nguyen, L. Xie, Single landmark distance-based navigation, *IEEE Trans. Control Syst. Technol.* 28 (5) (2019b) 2021–2028.
- [15] B. Zhu, A.H.B. Zaini, L. Xie, Distributed guidance for interception by using multiple rotary-wing unmanned aerial vehicles, *IEEE Trans. Ind. Electron.* 64 (7) (2017) 5648–5656.
- [16] M. Cao, C. Yu, B.D.O. Anderson, Formation control using range-only measurements, *Automatica* 47 (4) (2011) 776–781.
- [17] Y. Zhou, Q. Zhang, H. Wang, P. Zhou, T. Chai, EKF-based enhanced performance controller design for nonlinear stochastic systems, *IEEE Trans. Automat. Control* 63 (4) (2017) 1155–1162.
- [18] T. Bailey, J. Nieto, J. Guivant, M. Stevens, E. Nebot, Consistency of the EKF-SLAM algorithm, in: *2006 IEEE/RSJ International Conference on Intelligent Robots and Systems*, IEEE, 2006, pp. 3562–3568.
- [19] A. Barrau, S. Bonnabel, Extended Kalman filtering with nonlinear equality constraints: a geometric approach, *IEEE Trans. Automat. Control* 65 (6) (2019) 2325–2338.
- [20] R. Cen, X. Zhang, Y. Tao, F. Xue, Y. Zhang, Temporal delay estimation of sparse direct visual inertial odometry for mobile robots, *J. Franklin Inst.* 357 (7) (2020) 3893–3906.
- [21] C. Paleologu, J. Benesty, S. Ciochina, A robust variable forgetting factor recursive least-squares algorithm for system identification, *IEEE Signal Process. Lett.* 15 (2008) 597–600.

- [22] P. Ioannou, B. Fidan, Adaptive control tutorial, in: SIAM, 2006.
- [23] P.A. Ioannou, J. Sun, Robust Adaptive Control, Courier Corporation, 2012.
- [24] K. Guo, X. Li, L. Xie, Ultra-wideband and odometry-based cooperative relative localization with application to multi-UAV formation control, IEEE Trans. Cybern. 50 (6) (2019) 2590–2603.
- [25] J. Wang, E. Olson, AprilTag2: efficient and robust fiducial detection, in: Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2016.